

Pedro Silva

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EDUCATION

Instituto Superior Técnico, University of Lisbon

MSc in Computer Science and Engineering

Lisbon, Portugal

Sep. 2025 – Aug. 2027 (expected)

Faculty of Sciences, University of Lisbon (FCUL)

BSc in Informatics Engineering

Lisbon, Portugal

Sep. 2022 – Aug. 2025

EXPERIENCE

Software & AI Team Member

2025 – Present

AeroTec ATLAS, Systems & Aeronautics (IST UAV team)

Lisbon, Portugal

- Built modular Python/GStreamer video pipelines with ROS 2 integration for drone capture, processing and streaming
- Developed onboard and ground-station modes for low-latency WifiBroadcast video, with RTSP backup over ZeroTier and independent MAVLink control
- Integrated RealSense-based ORB-SLAM3 and YOLO person detection through DeepStream/TensorRT on Jetson, with supervision and restart of failed processes

Programming Content Creator

2021 – Present

YouTube channel zFake

youtube.com/@zFake

- Teach programming and computer science to a Portuguese-speaking audience, including a 2D game development series in vanilla JavaScript with open-source companion code

Avionics Team Member, PalmaSat

2022

CanSat Portugal, 9th edition (ESERO)

Portugal

- Co-built the flight software of a can-sized microsatellite (telemetry, sensors, radio downlink); awarded the national **Technical Performance Award**

PROJECTS

Sophia | TypeScript, discord.js, SQLite, Docker | github.com/zF4ke/sophia

- Built a self-hosted Discord AI agent for multi-step research, sandboxed code execution and file/media analysis, with server-history and web search
- Implemented persistent tasks with mid-task steering, saved-work resumption, scheduled follow-ups and memory, with permission-controlled actions and Docker workspaces

Ada | TypeScript, Electron, SQLite, knowledge graphs | Private

- Built a personal AI working partner and second brain that connects notes, documents, preferences and task experience in a persistent knowledge graph and uses accumulated knowledge and feedback to guide future tasks
- Developed an extensible task runtime for research, coding and application building. Ada can create missing tools, use them to complete ongoing tasks, and modify its own code, prompts and capabilities through evaluated builds

Timebox | TypeScript, Electron, multi-agent LLM systems, Vitest | github.com/zF4ke/timebox

- Built a desktop scheduler where five specialist AI agents critique and vote on weekly plans under configurable quorum and iteration limits, with calendar (.ics) export
- Designed a benchmarking harness that runs a model/quorum/iteration matrix with pre-flight cost estimates, budget caps, per-call usage traces and deterministic mistake scoring

ModernBazaar | Java, Spring Boot, TypeScript, Next.js, PostgreSQL | github.com/zF4ke/ModernBazaar

- Built a Hypixel SkyBlock market analytics platform with Spring Boot ingestion, Next.js dashboards and PostgreSQL history; added Docker deployment, Prometheus/Grafana monitoring, Auth0 login and Stripe subscriptions

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Electron, Spring Boot, Tailwind CSS

AI & Memory: LLM agents, tool calling, knowledge graphs, semantic retrieval, PyTorch

Robotics & Vision: ROS 2, OpenCV, GStreamer, DeepStream, TensorRT, ORB-SLAM3, MAVLink

Databases: PostgreSQL, SQLite, MongoDB, Firebase

Developer Tools: Git, Linux, Docker

Spoken Languages: Portuguese (native), English (fluent)